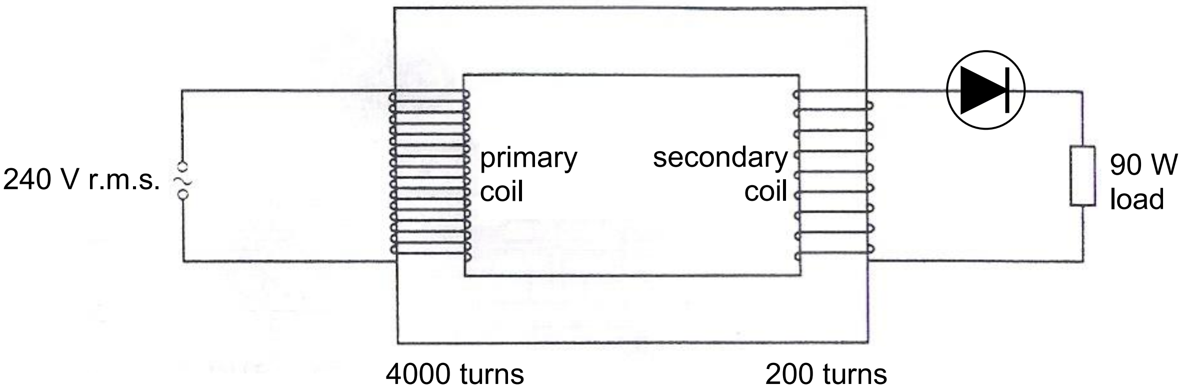
28 The diagram shows an iron-cored transformer assumed to be 100% efficient. The primary coil of the transformer has 4000 turns and is connected to a 240 V r.m.s. supply. The secondary coil has 200 turns and is connected, through an ideal diode, to a resistive load which is dissipating energy at a mean rate of 90 W.



What is the r.m.s. current in the secondary coil.

A 0.375 A **B** 0.750 A **C** 7.50 A **D** 10.6 A

Calculate the Vs : Vs(rms) = 200/4000 x 240 = 12 V

$(12)^2$ / R = 180 W Hence R = 0.80 Ω . The power is 180 W as the diode remove half the power.

I^2 (0.80) = 90 W ,

I = 10.6 A
Ans: D

29 A metal surface is illuminated with a beam of monochromatic electromagnetic radiation. By the